

Data Visualization

Fall 2026

Fridays 9am - 11am

Avery 409

Week 3

1. Found visualizations
2. Introduction to code
3. Assignments

Found Visualizations

Introduction to code

Our Work space

Making

Text editor - ie visual studio code

Viewing and debugging

Local server and browser

Or Liveserver within visual studio code

Sharing and collaborating

Github pages

Languages

Markup language: HTML

Stylesheet language: CSS

Coding language: JavaScript

A coding language involves logic

What is software?

A set of instructions, data or programs used to operate computers and execute specific tasks

What is an algorithm?

A process, procedure, or set of rules that must be followed to solve a problem

What is a coding language?

A formal set of words, symbols, and rules used to write instructions that a computer can understand and carry out

The Online Historical Encyclopaedia of Programming Languages (HOPL)
<https://hopl.info/>

8,945: recorded languages in history (HOPL)

~700: notable enough to be documented

~50: in regular real-world use

10–20: the core group where most code is written today

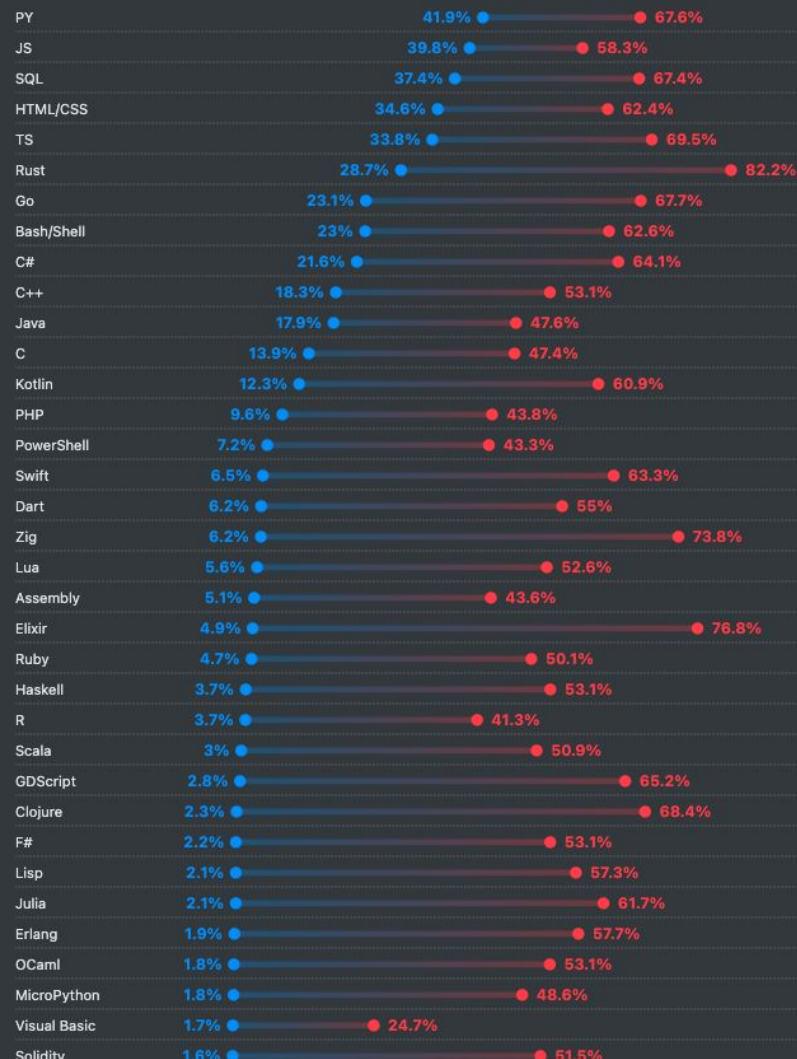
Language popularity

Programming, scripting, and markup languages

JavaScript, Python and SQL are all highly-desired and admired programming languages, but Rust continues to be the most-admired programming language with an 83% score this year.

2 Which **programming, scripting, and markup languages** have you done extensive development work in over the past year, and which do you want to work in over the next year? (If you both worked with the language and want to continue to do so, please check both boxes in that row.)

<https://survey.stackoverflow.co/2024>



Language popularity

“The ratings are based on the number of skilled engineers world-wide, courses and third party vendors. Popular web sites Google, Amazon, Wikipedia, Bing and more than 20 others are used to calculate the ratings. It is important to note that the TIOBE index is not about the *best* programming language or the language in which *most lines of code* have been written.”

Sep 2026	Sep 2025	Change	Programming Language		Ratings	Change
1	1			Python	17.76%	-8.22%
2	3	⬆		C	10.28%	+1.63%
3	2	⬇		C++	8.67%	-0.13%
4	4			Java	7.54%	-0.81%
5	5			C#	4.22%	-2.16%
6	6			JavaScript	2.76%	-0.46%
7	7			Visual Basic	2.55%	-0.28%
8	11	⬆		SQL	2.16%	+0.29%
9	13	⬆		R	1.69%	+0.27%
10	18	⬆		Rust	1.34%	+0.33%

High and low

human-like
(abstract, readable, far from hardware)



most machine-like
(close to what the CPU actually runs)

Python (1991): AI, data science, automation
`print("Hello, World!")`

JavaScript (1995): Websites, web servers
`console.log("Hello, World!");`

PHP (1995): Web pages, WordPress
`<?php echo "Hello, World!"; ?>`

Perl (1987): Text processing, sysadmin scripts
`print "Hello, World!\n";`

Java (1995): Android apps, backend services
`System.out.println("Hello, World!");`

C# (2000): Windows apps, Unity games
`Console.WriteLine("Hello, World!");`

Erlang (1986): Messaging and chat servers
`io:format("Hello, World!~n").`

C++ (1985): Game engines, high-performance apps
`std::cout << "Hello, World!\n";`

C (1972): Operating systems, embedded devices
`printf("Hello, World!\n");`

Assembly (late 1940s): Drivers, bootloaders
`mov rax, 1 / mov rdi, 1 / mov rsi, msg / mov rdx, 13 / syscall`

Machine Code (1940s): What the CPU runs directly
`B8 01 00 00 00` (just the "put 1 in a register" step on its own)

Programming

Logic

a system or set of principles underlying the arrangements of elements in a computer or electronic device so as to perform a specified task.

Syntax

the arrangement of words and phrases to create well-formed sentences in a language.

Programming languages share logic not syntax

Code Basics The most important component

You can tell how well code is organized from across the room. Or by squinting or zooming out. The shape of code from 20 feet away is incredibly informative. Clean code is idiomatic, as brief as possible, obvious even if it's not heavily documented. Colloquial and friendly. As was written in *Structure and Interpretation of Computer Programs* (aka SICP), the seminal textbook of programming taught for years at MIT, "A computer language is not just a way of getting a computer to perform operations ... it is a novel formal medium for expressing ideas about methodology. Thus, programs must be written for people to read, and only incidentally for machines to execute." A great program is a letter from current you to future you or to the person who inherits your code. A generous humanistic document.

Machines now also read comments!!!

3 parts What are html, css, and javascript?

An analogy:

HTML ... defines the actors on stage: how many, what roles.

CSS ... moves the actors to different positions, and changes their costumes.

JS ... is the theater director. JS can modify HTML and CSS: can add or remove actors, change their roles, costumes, positions.

JS can also communicate with both the audience (you, or your mouse) and the back-stage crew (the rest of the browser.)



From
Agnes!

<https://www.nytimes.com/by/agnes-chang>

Code Basics Who reads our code?

There are 3 “audiences” for our code,

1. The machine
2. Me
3. Your future self

Therefore:

Stay organized

Structure it well

Name variables and functions coherently

Comment and document generously

Code Basics commenting your code

HTML *<!-- Write your comments here -->*

Usually when something is commented out, your text editor will turn the text gray

CSS */* Write your comments here */*

JS *// Write your comments here*

```
.....div.each(function(){  
.....var div = d3.select(this),  
.....g = div.select("g");  
  
.....// Create the skeletal chart.  
.....if (g.empty()) {  
.....div.select(".title").append("a")  
......attr("href", "javascript:reset(")  
......attr("class", "reset")  
......text("reset")  
.....}
```

HTML basics

DOM

Document Object Model refers to the hierarchical structure of HTML

Let's take a look at a basic page!

go to any web page (I'll use google homepage) on your computer in Chrome.

```
Open html ----- <html>
Open body ----- <body>
Open header1----- <h1>Breaking News</h1> ----- Close header1
Open paragraph ----- <p></p> -----Close paragraph
Close body----- </body>
Close html----- </html>
```

Your browser

Is a powerful tool

Inspect Element

Welcome

Planning investigations >

Checklist

Build your own datasets >

Undocumented APIs

Browser automation

Best practices

Appendices >

References

Inspect Element

A Practical Guide to Data Investigations

This guide walks through in-depth case studies and hands-on tutorials to help you investigate opaque systems, systematically.

You will learn how to build your own datasets by [finding undocumented APIs](#), [automating browsers](#), and crowdsourcing. Most importantly, you'll build the intuition to develop a [hypothesis](#), and bullet-proof methodologies to answer key questions for your reporting and research.

As practitioners, we've had to learn on the job, hit dead ends, defend our decisions, and seek external expertise across fields and industries. Here we distill these experiences and include tips for veterans and new-comers alike.

The disciplines we'll draw from include: investigative journalism, data engineering, computer science, social science, and other branches of information science.

Don't code? No problem: the guide emphasizes underlying principles and uses plain-language ("pseudocode") explanations to accompany any code.



<https://inspsectelement.org/>

HTML basics

Text editor

```
1  <!DOCTYPE html>
2  <html lang="en">
3  ▼  <head>
4      <meta charset="utf-8">
5      <title>Page Title</title>
6  ▲  </head>
7  ▼  <body>
8      
9      
10 ▲  </body>
11  </html>
```

Nested structure

Open html declare the language is english

Open head tag

declare character encoding is utf-8

Open title tag Page Title *close title tag*

Close head tag

Open body tag

Close body tag

Close html

HTML basics

Nested structure

HTML code

```
<body>  
  <strong>  
    this text is now bold  
  </strong>  
  <i>  
    this text is now italic  
  </i>  
</body>
```

description

```
Open body tag  
  Open bold  
    Bolded text  
  Close bold  
  Open italic  
    Italicized text  
  Close italic  
Close body tag
```

HTML basics

Nested structure

HTML code

```
<body>  
  <strong>  
    <i>  
      this text is now bold and italic  
    </i>  
  </strong>  
</body>
```

description

```
Open body tag  
  Open bold  
    Open italic  
      Bolded and italicized text  
    Close italic  
  Close bold  
Close body tag
```

HTML basics

Space and Line returns

A space(hitting the spacebar) doesn't mean the same thing in code.

White space does not matter in html, css or js code

To actually produce a space or a line break in your text on you HTML page, you need one of these

HTML

fixed space
Hard space
NBSP (non-breaking
space)

** **

Line break
Carriage return

**
**

Paragraph open and close
tags produce automatic
line break

<p> </p>

CSS basics

P is the selector for paragraphs

What does this code do?

```
p {  
    background-color: #aaaaaa;  
}
```

Selector open curly brace

Category of style color specific style semicolon

Close curly brace

CSS basics

*These all work... they are all the same
...but which is the easiest to read?*

```
p {  
    background-color: #aaaaaa;  
}
```

```
p {  
background-color: #aaaaaa;  
}
```

```
p {background-color:#aaaaaa;}
```

```
p {  
  
background-color: #aaaaaa;}
```

CSS basics

CSS defines the style of the content between the tags on your page.

CSS applies to ALL instances of that tag.

HTML

```
<p>My paragraph  
blah blah  
blah</p>
```

CSS

```
p{  
    color: red;  
    font-family: serif;  
}
```

The page you see

My paragraph
blah blah blah

CSS basics

CSS defines the style of the content between the tags on your page.

CSS applies to ALL instances of that tag.

HTML

```
<p>My paragraph blah blah blah</p>
<p>another paragraph
</p>
<p>another paragraph
</p>
<p>another paragraph
</p>
```

CSS

```
p{
    color: red;
    font-family:
    serif;
}
```

The page you see

My paragraph blah
blah blah

another paragraph

another paragraph

another paragraph

Naming Conventions

Rules:

1. Cannot start with a number
2. Cannot contain a space
3. Cannot have special characters
4. Cannot be reserved words/keywords
5. Case sensitive

Common styles:

camelCase

under_score

HTML/CSS/JavaScript Selectors: Classes and IDs

HTML

Classes can be applied to multiple divs

Ids can only be applied to 1

```
<div id="A">
```

Text with id of A

```
</div>
```

```
<div class="A">
```

Text with class of A

```
</div>
```

CSS

Classes use dot to select .A

Ids use # to select #A

```
#A{
```

font-size: 24px;

```
}
```

```
.A{
```

color:red;

```
}
```

HTML/CSS Selectors

HTML

*Classes can be applied to multiple divs
Ids can only be applied to 1*

```
<body>
  <div id="A">
    <p>Text with id of A</p>
  </div>
  <div class="A">
    <p>Text with class of A</p>
  </div>
  <div class="A B">
    <p>Text with class A and class B</p>
  </div>
</body>
```

CSS

*Classes use dot to select .A
Ids use # to select #A*

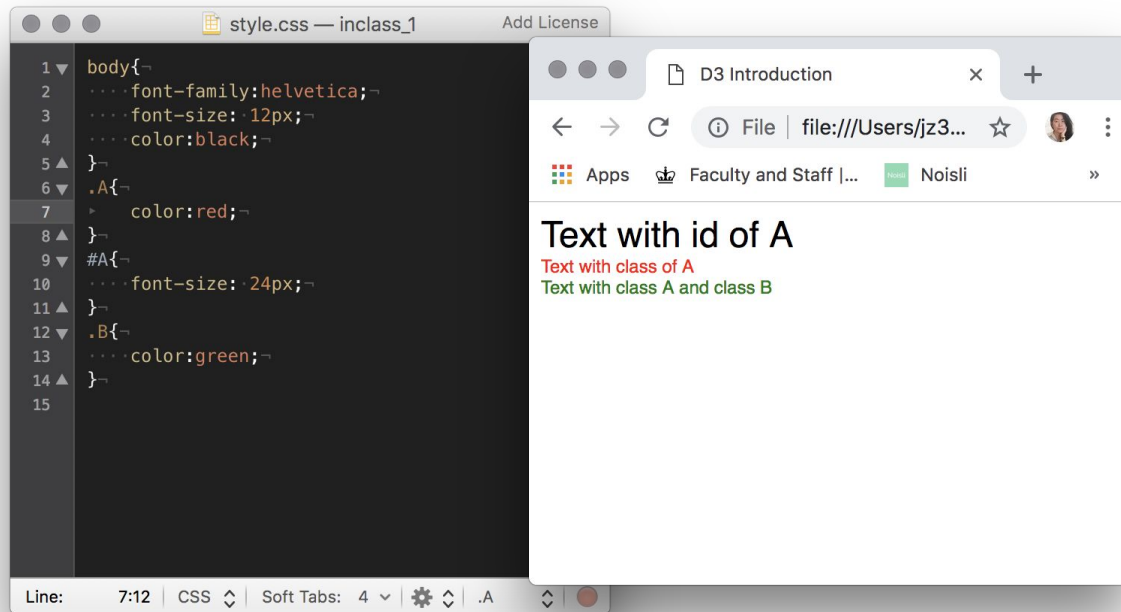
```
body{
  font-size: 12px;
  color: black;
}
.A{
  color: red;
}
#A{
  font-size: 24px;
}
.B{
  color: green;
}
```

What we see on the browser
page

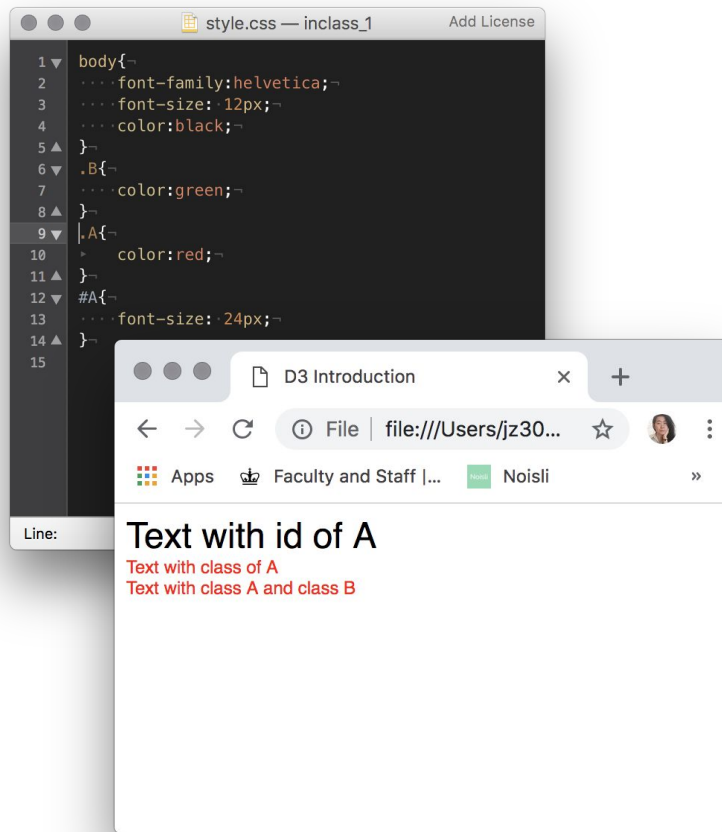
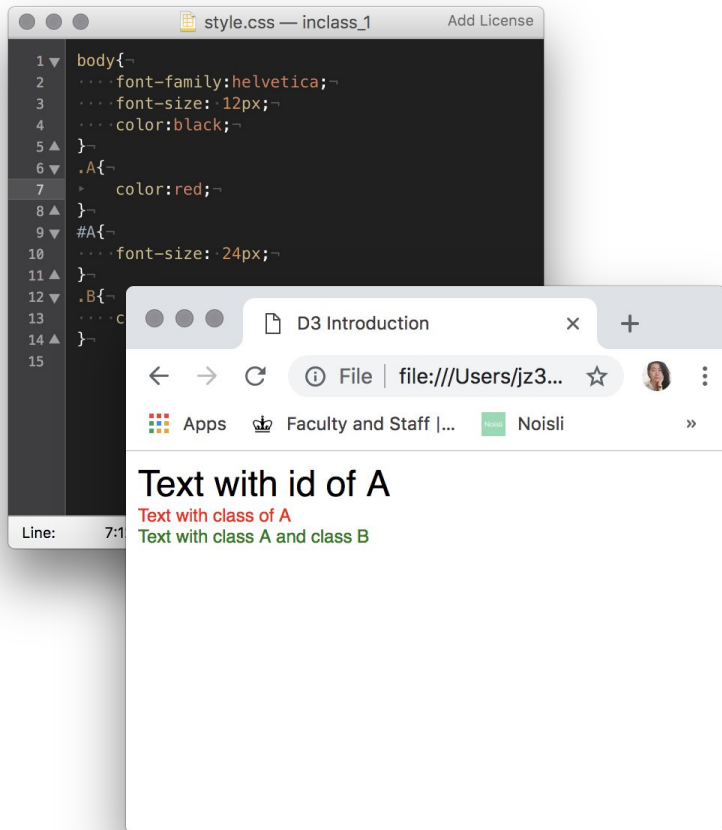
?

HTML/CSS Selectors

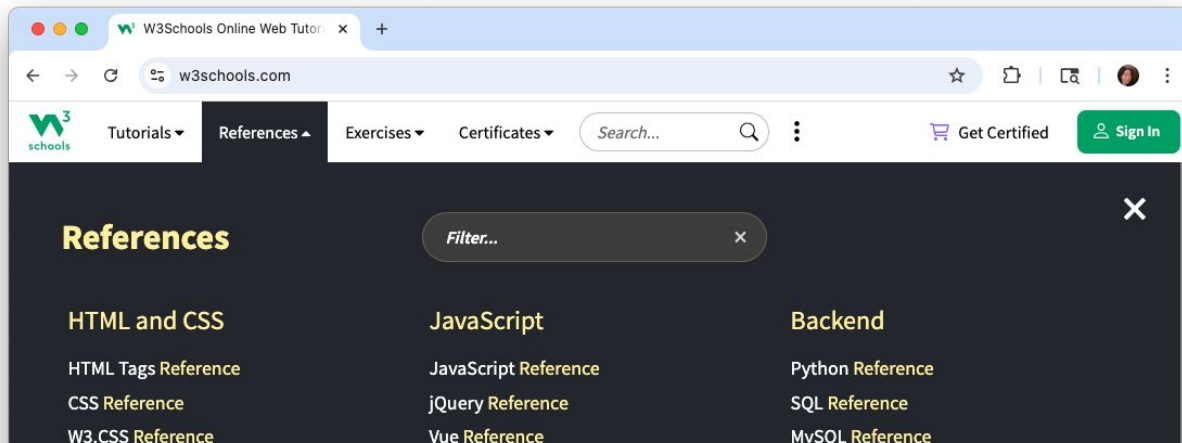
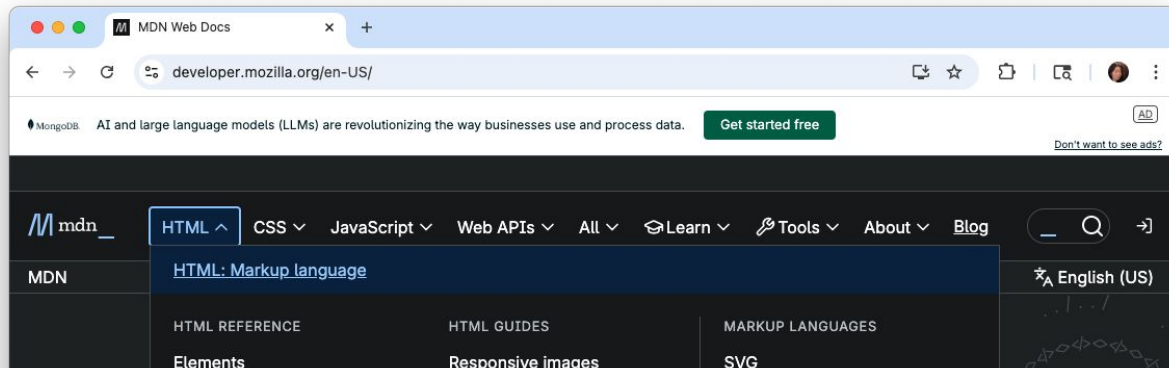
```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>D3 Introduction</title>
    <script src="d3/d3.js"></script>
    <link rel="stylesheet" type="text/css" href="style.css">
  </head>
  <body>
    <div id="A">
      Text with id of A
    </div>
    <div class="A">
      Text with class of A
    </div>
    <div class="A B">
      Text with class A and class B
    </div>
```



HTML/CSS Selectors



HTML/CSS References



HTML is drawn in the order it is written from top to bottom.

CSS is executed in the order it is written from top to bottom.

Javascript is not.

Javascript Basics

components of code

Things

Primitives

Structures

The instructions we give things

Conditional statements

Operators

Javascript Basics Variables

3 keywords for variables

var - generic

let - can be changed

const - cannot change its value

Using variables

Declare

var a = 2

var b = 3

Use

a+b

Change

a = a+1

outputs 5

a is now 3

Javascript Basics Primitives

https://en.wikipedia.org/wiki/Primitive_data_type

These are built in to JavaScript and we will use them:

Integer - 1, 2, 3, 2000000 , -30

Float - decimal number, 0.2, -5,4

Boolean - *true, false*

String - *"any kind of sentence or word" , "20"*

Character - a character like "a", "!"

Javascript Basics Primitives

*1.0 is a **float***

$1.0 + 1 = 2.0$

*1 is an **integer***

"1" + "1" = "11"

*"1" is a **string***

$"1" + 1 = ?$

Javascript Basics Primitives

Open your browser console and try it out

```
> "1"+1
```

```
< "11"
```

```
> 1+"1"
```

```
< "11"
```

```
> 1+1.0
```

```
< 2
```

```
> 1+parseFloat("1")
```

```
< 2
```

```
> "1"*1
```

```
< 1
```

```
> Math.round("1")
```

```
< 1
```

Javascript Basics Data Structures

2 ways we organize sets of values

arrays = lists []

objects = dictionaries { }

Javascript Basics Arrays

Javascript syntax:

[value, value, value]

Square bracket *value comma value comma*
value square bracket

Javascript Basics Arrays

Some examples:

`[1, 2, 3, 4]`

list of integers

`["1", "2", "3", "4"]`

list of strings

`[1, "red", "2", 4.3]`

mixed list

`[1,1,1,1]`

list of identical values

`[[1, 2, 3, 4], ["1", "2", "3", "4"], [1, "red", "2", 4.3], [1,1,1,1]]`

list of lists

Javascript Basics Arrays

Lists are linear:

They have an index - the order they come in.

The index starts at 0

Because they have an order, you can sort lists.

They also have a length

Javascript Basics Objects

Javascript syntax:

{Key:value, key:value}

Curly bracket *key colon value comma key colon value curly bracket*

Javascript Basics Objects

Some examples:

```
{ "blue": 56, "black": 20 }
```

```
{ "1": "red", "2": "green" }
```

```
{ "25": 2, "a": 2, "blue": 90 }
```

```
{ "geoid2": 25, "geoid4": 40 }
```

✗ NO!!!!!!

```
{ "geoid3": 2, "geoid3": 90 }
```

***Objects must have
UNIQUE keys***

Javascript Basics Objects

Objects are not linear:

- no index - no first item of object

- Cannot be sorted as object - You can sort by key or value into an array.

- There is no length - you can get length of all the keys or values.

- JSON - JavaScript Object Notation

Javascript Basics Arrays and Objects

```
var k = [5, 2, 6]
```

To get the first item in the array called "k" above in javascript: `k[0]`

What is the value of `k[1]`?

How do we find the value 6?

What is the index of 2?

```
var j = {"A":1, "B":2, "C":4}
```

To get the a given item in the object called "j" above in javascript, you would need to know the key.

`j["A"]` has the value of 1

What is the index of the value 2?

Javascript Basics Arrays and Objects

Array of arrays	<code>[[1,2,3] , [2,5] , [5,9]]</code>
Array of objects	<code>[{a:2} , {b:3} , {c:2,d:2}]</code>
Object containing objects	<code>{ a:{j:2,k:4}, b:{j:2,k:12}}</code>
Object containing arrays	<code>{ a:[1,2,3], b:[3,4,5] }</code>

Javascript Basics Arrays and Objects

excel

A1			
	A	B	
1	Food	Deliciousness	
2	Apples	9	
3	Green Beans	5	
4	Egg Salad Sa	4	
5	Cookies	10	
6	Liver	0.2	
7	Burrito	7	
8			

Comma Separated Values
CSV
in text editor

```
Food,Deliciousness
Apples,9
Green Beans,5
Egg Salad Sandwich,4
Cookies,10
Liver,0.2
Burrito,7
```

JavaScript Object Notation
JSON
Object

Parsed as array

```
[{Food:"Apples", Deliciousness:9},
{Food:"Green Beans",Deliciousness:"5"},
{Food:"Egg Salad
Sandwich",Deliciousness:"4"},
{Food:"Cookies",Deliciousness:"10"},
{Food:"Liver",Deliciousness:"0.2"},
{Food:"Burrito",Deliciousness:"7"}]
```

Javascript Basics

components of code

Things

Primitives

Structures

The instructions we give things

Operators

Conditional statements

Javascript Basics Operators

Math Operators

+	add
-	subtract
*	multiply
/	divide
%	mode - the remainder

Comparison Operators

>	greater than
<	less than
>=	is greater than or equal to
<=	is less than or equal to
==	is equal to
===	is equal to (in value as well as in <u>type</u>)

Logical Operators

&&	and
	or
!	not

1+3	4
"1" + "3"	"13"
"This" + "class"	Thisclass
"" + title + ""	title

7/2 3.5

7%2 1 Try a few examples

1 == "1"	true
1 === "1"	false

- if we want to have a number that is divisible by 3 AND 5
- if we want to have a number that is divisible by 3 OR 5
- if we want to have a number that is divisible by 3 but NOT 5

Javascript Basics Logic

3 Conditional Statements

Conditional statements test the state of your variables with a condition, if it satisfies that condition, the program does something, if not, then it does something else.

```
If(condition){  
    Do this  
}  
else{  
    Do this  
}
```

```
If x is divisible by 5  
    Print out "ok"  
If it is not  
    Print out "not ok"
```

```
If(x%5==0){  
    console.log("ok")  
}  
else{  
    console.log("not ok")  
}
```

```
For(initial state, limit, change){  
    Do this  
}
```

As long as variable i is between 0 and 100
Print out i

```
For(var i = 0; i<100; i++){  
    console.log(i)  
}
```

```
While(condition){  
    Do this  
}
```

As long as variable i is between 0 and 100
Print out i

```
Var i = 0  
while(i<100){  
    console.log(i)  
    i=i+1  
}
```

Javascript Basics Functions

```
function nameOfFunction(input)

{
    Output = Some operations with input
    return output
}
```

return

```
function addThree(input)

{
    return input + 3
}
```

addThree(3)

6

addThree(22)

25

Demo 1 logic

Problem: Find all the numbers between 1 and 100 that are divisible by 3, divisible by 5, or divisible by both.

Pseudocode - group exercise

1. **Print out all the numbers between 1 and 100**
- 2.

For Next Week

Visual reverse-engineering assignment

Next Week

Data

Journalism - long and short forms

Creative Agencies and Advertising

Individual Authors and Creators

Awards and Conferences

Academic Research

Journalism

Some Examples:

NYT

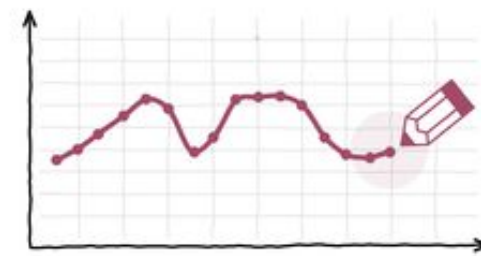
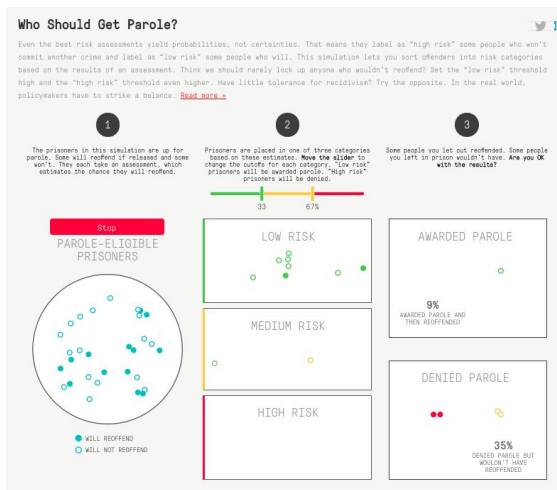
Washington Post

The Guardian

LA Times - video

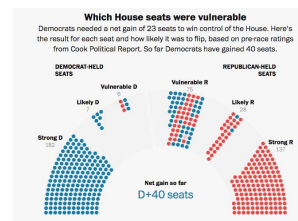
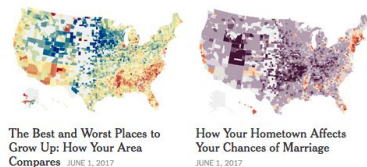
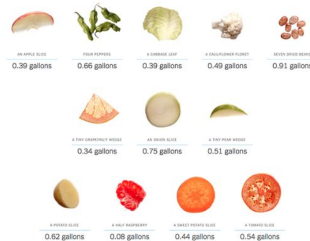
Propublica

Economist

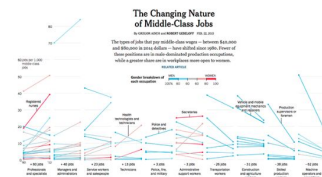


You Draw It: How Family Income Predicts Children's College Chances MAY 26, 2016

Infographic: Your Contribution to the California Drought
By Larry Buchanan, Josh Keller, and Haeyoun Park



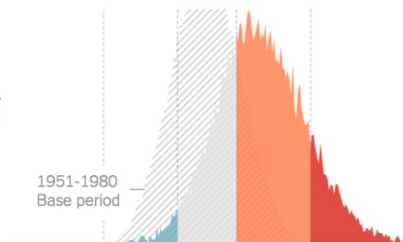
In Data Visualisation: The Changing Nature of Middle-Class Jobs
by Gregor Aisch and Robert Gebeloff



It's Not Your Imagination. Summers Are Getting Hotter.

JULY 28, 2017

Summer temperatures have shifted toward more extreme heat over the past several decades.



2020: The Year in Visual Stories

nytimes.com/interactive/2020/12/30/us/2020-year-in-graphics.html

The New York Times

In a year with so many world-shaking moments, our strongest visual stories covered impeachment, outbreak, caucuses, primaries, donations, delegates, shutdown, jobs, deaths, coughing, hospitals, testing, George Floyd, protests, tear gas, dreams, nursing homes, policing, the election, stimulus, symptoms, Beirut, power, the Supreme Court, ballots, wildfires, Trump's taxes, turnout, swing states, masks, results, A.I., voter shifts, I.C.U.s, pollution and vaccines.

2020: The Year in Visual Stories and Graphics.

By The New York Times December 30, 2020

2023: The Year in Visual Stories

nytimes.com/interactive/2023/us/2023-year-in-graphics...

The New York Times

ARTICLES FREE TO READ FOR A LIMITED TIME

This year, our visual stories covered a range of subjects: artificial intelligence, war in Ukraine and Gaza, the transition to cleaner energy, natural disasters and the 50th anniversary of hip-hop. **2023: The Year in Visual Stories and Graphics.**

By The New York Times Dec. 26, 2023

January February March April May June July August September October November December

January

What We Know About the Gun Used in the Monterey Park Shooting

Dancing With the Stairs

Six N.F.L. Players Show Football's Grind on Their Feet and Hands

Dissecting Elon Musk's Tweets: Memes, Rants, Private Parts and an Echo Chamber

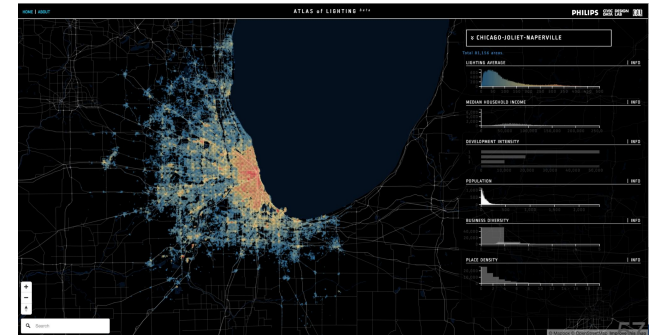
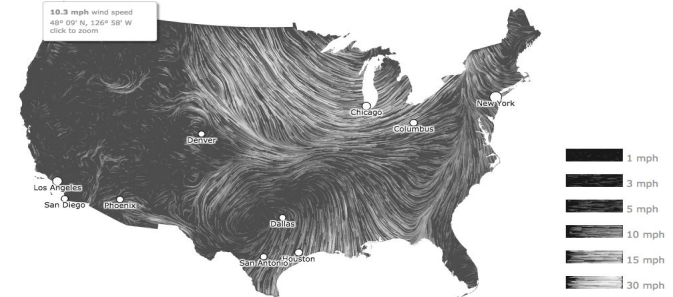
"After Elon Musk bought Twitter in November 2022, there was a real sense of urgency to save as many tweets as possible. You didn't really know what could happen the next day. After going through the data, we decided to start

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Research

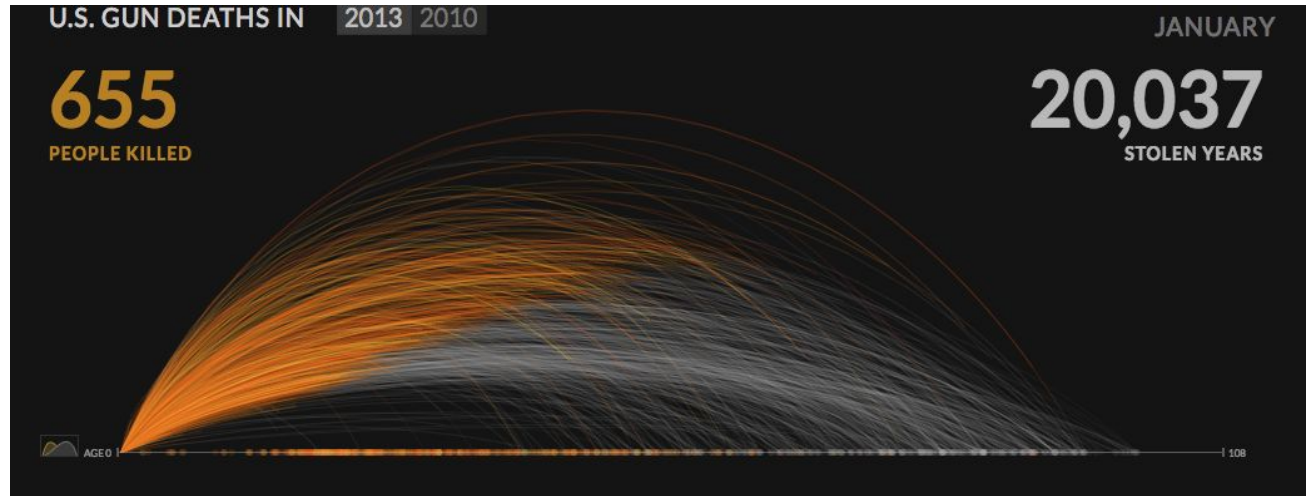
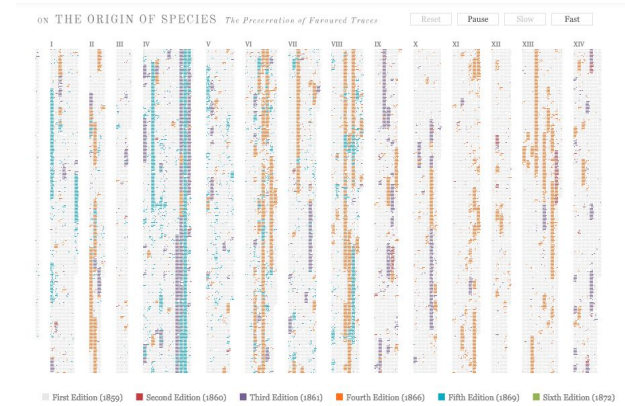
Center for Spatial Research

Uber
Google creative lab
Senseable Cities
UW Interactive Data Lab
Visual Computing Group
Civic Data Design Lab



Agencies

Pentagram
Fathom
Periscopic
Stamen



People

<http://www.richardthe.com/>

<https://ncase.me/>

<https://www.visualcinnamon.com/about>

<http://giorgialupi.com/about/>

<http://www.radicalcartography.net/>

<http://felinlovewithdata.com/>

<http://truth-and-beauty.net/>

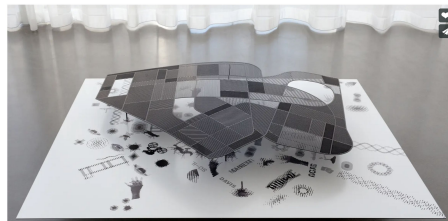
<https://flowingdata.com/about-nathan/>

Richard The

Work About

Manetti Shrem Museum of Art

For the opening of the Manetti Shrem Museum of Art at UC Davis we worked with [So-Il](#) on an exhibition about the concepts behind the architecture, called Museum as Process. It featured photographs by [Luisa Lambri](#), and two pieces from us, [Connective Field](#) and [The Zone](#).



Nicky Case

blog | twitter | throw monies at me

• THINGS YOU CAN PLAY •

EXPLORABLE EXPLANATIONS

explorable explanations, a hub for learning through play



the evolution of trust, an interactive guide to the game theory of why & how we trust each other



how to remember anything forever-ish, an interactive comic on the science of memory



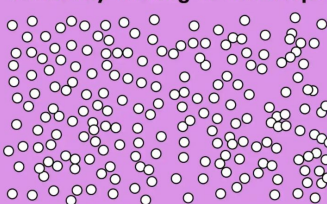
parable of the polygons, a playable post on the shape of society

Life and style

How to understand Covid-19 antibody testing in 10 steps

Inaccuracies in the testing, such as false negatives and false positives, are potentially harmful

How To Understand COVID-19 Antibody Testing in Ten Steps



OK so first off, imagine 200 people who *think* they might have been infected. (See the circles above? That's them!)



Giorgia Lupi is an information designer.

She is the co-founder and Design Director of [Accurat](#), a design studio that works with clients like [Google](#) and [Twitter](#).

Her work is part of the permanent collection of the [Museum of Modern Art](#), where in 2017 she also was commissioned to create a large-scale site-specific piece.

Her TED Talk on her humanistic approach to data has over 1 million views.

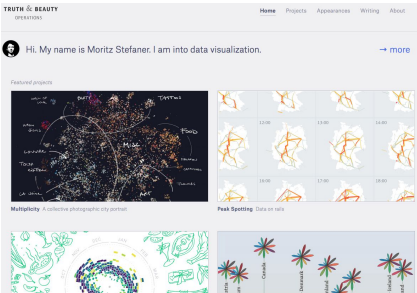
An Astronomer passionate about effective & engaging data visualization



Hi my name is [Alexandru](#), which is pronounced like the first, an origin about that of the word "Alexandru" is from the Greek word "Alexandros" which means "defender of men".

I'm a practicing data visualization designer and artist, working from around the world with [Google](#) and [Twitter](#). I've also been a speaker at [DataCamp](#) and [DataCamp](#).

I made this video to figure out that, although I love doing data projects, I don't have the visualization skills and insights. In my evenings and weekends I started working myself about the subject of data visualization and data science learning journey. I hope you'll find the opportunity to join [Adapt](#) as a full-time data visualization designer. Thanks to you all.



RADICAL CARTOGRAPHY

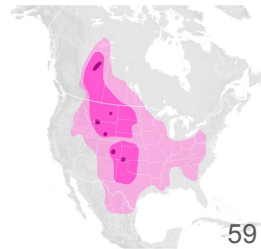
ABOUT PROJECTS

BROKEN BY GEOGRAPHY
BROKEN BY PROBABILITIES
BROKEN BY CHANCE

GEOMETRY AND TOPOLOGY
VIBES MAPS
HUMAN GEOGRAPHY
AMERICAN HISTORY
GENETIC DEMOGRAPHICS

Use all projects appear in each group. Some projects appear more than once.

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Events



The premier forum for
advances in visualization and
visual analytics



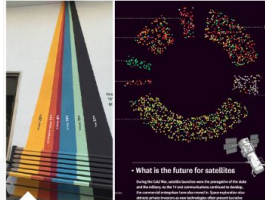
[About](#) [News](#) [Awards](#) [Entry Showcase](#) [Sponsor](#) [Attend the Awards ceremony](#)



ABOUT 1 MONTH AGO

Dear Winners: Your Trophy Awaits

Are you a 2018 Information is Beautiful Awards Winner? Would you like to buy a duplicate trophy? We've got you - send us an email ASAP with your details, and we will make arrangements to ... →



ABOUT 2 MONTHS AGO

The Winners: Special Awards

These prizes were established to honor the students, prolific individuals, non-English outfits, and other notable people whose work received top votes in this year's Awards. →

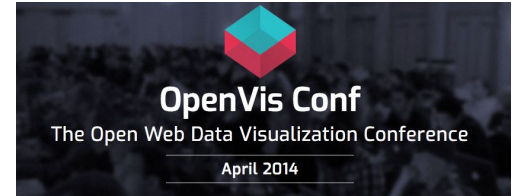


ABOUT 2 MONTHS AGO

The 2018 Awards Ceremony

We sipped bubbly beverages, mingled with dataviz celebrities, noshed on Shortlist Cookies, and enjoyed a memorable evening honoring fantastic designers from all over the world. →

Even the older versions of these 2 conferences are great!



VIDEOS

MIKE BOSTOCK Design is a Search Problem	+	
ERIC FISHER Mapping Billions of Dots	+	
SAM BELIKOFF Using D3 with Backbone, Angular and Ember	+	
KENNEDY_ELLIOT kennedy_elliot	+	
JEN CHRISTIANSEN Visualizing Science: Developing Information Graphics for Scientific American Magazine	+	

